

CHAINLAB

Solana Development Concepts

Programs, accounts, testing and deployment workflow

FREE BLOCKCHAIN EDUCATION RESOURCE

This guide is designed for learning and revision. It explains foundational concepts and highlights practical limitations and safety considerations.

Learning objectives

- Understand the stages of building a Solana program.
- Recognize account validation and testing responsibilities.
- Plan a safe deployment workflow.

1. Program design

- Start by defining instructions, accounts, authority rules, state transitions and failure cases. Keep the scope small enough to test thoroughly.
- For each instruction, document which accounts are read or written, who must sign, and which invariants must always hold.

2. Account validation

- Programs must validate account ownership, signer status, writable status, expected addresses, data layout and relationships between accounts.
- Never assume that a frontend has correctly validated inputs. Enforce security-critical checks in the program itself.

3. Development workflow

- Create a reproducible project and pin toolchain/dependency versions where practical.
- Implement one instruction at a time and add tests for normal behavior, invalid inputs and unauthorized access.
- Use local testing and Devnet before a production release. Review logs and transaction failures rather than ignoring them.

4. Anchor framework

- Anchor provides conventions and tooling for Solana program development, including account constraints, instruction interfaces and generated client support.
- Framework constraints are helpful, but developers still need to understand what each constraint checks and what assumptions remain unverified.

5. Release checklist

- Review source and dependency changes; run automated tests; obtain independent security review appropriate to risk; verify deployment addresses; protect upgrade authority; publish clear user documentation; prepare monitoring and incident procedures.

Review questions

- Why must account checks happen on-chain?
- Give examples of negative test cases.

- What should be considered before retaining or relinquishing upgrade authority?

Educational notice: This material is for general education only. It is not financial, legal, tax or investment advice. Verify current network and project details before acting.